# Tunger Hong

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## EDUCATION

**Bachelor of Science** — *Carnegie Mellon University*
*Electrical and Computer Engineering* — *expected May 2027*

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Python, C++, C, x86 ASM, Matlab, System Verilog |
| **Graphics and Tools** | Vulkan, OpenGL, ImGUI, Blender |
| **Classes** | 15-472: Real Time Computer Graphics<br>15-418: Parallel Computer Architecture and Programming<br>15-463: Computational Photography<br>15-474: Neural Graphics |

## PROJECTS & EXPERIENCE

**Vulkan Rendering Engine**

- Built a general-purpose real-time 3D engine that handles various materials (environment, mirror, lambertian), exposure and tone mapping, and different types of lights (sun, sphere, and spotlights).
- Sustained under 10 ms per frame with 1,024 dynamic lights by batching light data into GPU storage buffers and specializing each shader to the scene's light count.
- Supported diffuse, environment-lit, and mirror materials through one shader, switching behavior per object without additional pipeline state.

**Real-Time Watercolor Stylization**

- Implemented a published real-time watercolor stylization technique.
- Built the GPU infrastructure as a three-pass GPU compute pipeline which made the shader more modular, allowing for stylization to be iterated on without touching Vulkan boilerplate.

**3D Drawing Application**

- Extended the engine below into a both a 3D sketch and paint tool by using the watercolor pipeline above.
- Fixed cursor alignment across high-DPI displays and letterboxed viewports, to avoid drawing surface and the window conflicts.

**Scotty3D**

- Implemented rasterization, mesh manipulation, pathtracing, and animation within Scotty3D, a 3D modeling, rendering, and animation package.

**SIGGRAPH** — *July 2026*
*Student Volunteer* — *Los Angeles, CA*

- Supported technical sessions as part of a volunteer team, managing room turnover and directing attendee flow between sessions.

## WORK EXPERIENCE

**Carnegie Mellon University** — *Jan 2026 - Present*
*Teaching assistant for 15-362 Computer Graphics* — *Pittsburgh, PA*

- Edited reciation material and lectured to a class of 50 students on Rasterization, Geometric Processing, Path Tracing, and Animation.
- Created introductory material and projects, and maintained multiple assignments and lesson guides for students.

**Boston Scientific** — *May 2025 - Aug 2025*
*Software Intern* — *Saint Paul, MN*

- Engineered and optimized data-processing pipelines for heart-failure monitoring and chemical-sensor studies, improving analysis throughput and reliability.
- Redesigned the file-storage architecture and previous Matlab library and replaced it with a robust Python library. Processing time halved from 58 seconds to 29 seconds for one day's data.
- Developed the backend foundation for an interactive data-visualization web application, designed for scalability and seamless integration with future database systems.